

Beanstalk BIP-39 Audit Summary

Prepared by Cyfrin

Version 1.2

Lead Auditors

Giovanni Di Siena

Carlos Amarante

Assisting Auditors

Dacian

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Beanstalk Farms on Beanstalk BIP-39. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number of findings by severity. It closes with Cyfrin's attestation of the review's scope.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 1.2 dated May 2, 2024, was delivered to Beanstalk Farms, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 26 days, the Cyfrin team conducted an audit on the Beanstalk BIP-39 code provided by Beanstalk Farms. The findings consist of 2 High, 4 Medium & 6 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Beanstalk BIP-39
Repository	Beanstalk
Commit	dfb418d185cd...
Audit Timeline	Oct 16th - Nov 20th, 2023
Methods	Manual Review, Unit Testing, Differential Testing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	–	–
High Risk	2	–	–
Medium Risk	4	–	–
Low Risk	6	–	–
Informational	9	–	–
Gas Optimizations	6	–	–
Total	27	–	–

This version of the report was issued before any fixes were reviewed: all findings were open.

6 Audit Scope

Cyfrin conducted an audit of Beanstalk based on the code present in the repository commit hash dfb418d, specifically the diff between this commit hash and 7606673 which largely pertains to the BIP-39 upgrade which introduces the Seed Gauge System for dynamic incentivization of deposits for different whitelisted tokens based on the ratio of deposited BDVs. Other changes introduced in the diff between these two commits, since the previous audit performed by Cyfrin, are also considered in the scope of this audit.

The scope also includes a script for the calculation of the BDV of unmigrated deposits at a given block, present in commit 120ae27, and code changes following EBIP remediations merged into original seed-gauge branch along with other BIP-39 remediations at commit hash 5142b8f.

7 Risk Classification

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Beanstalk is a permissionless algorithmic stablecoin protocol built on Ethereum. The protocol uses a novel dynamic peg maintenance mechanism to have the price of 1 BEAN (the Beanstalk stablecoin) continuously cross its peg value of 1 USD without centralization or collateral requirements.

The BIP-39 upgrade introduces the Seed Gauge System, which is designed to dynamically adjust the Grown Stalk distribution in each Season based on deposited BDV per whitelisted LP and Bean. To achieve this, the protocol considers four parameters when assessing the state of Beanstalk:

- **L2SR:** the USD value of the non-Bean assets trading against Bean that are whitelisted in the Silo, excluding the locked liquidity backing unripe assets.
- **Bean Price (deltaB):** the deficit or surplus of Bean in its liquidity pools, which determines whether the price is above, at, or below peg.
- **Change in Soil Demand:** a measure of the ability of Beanstalk to attract a sufficient number of creditors.
- **Pod Rate:** minted Pods less paid Pods, divided by the Bean supply.

As a result, the protocol considers 144 cases when updating the Temperature (interest paid for minting Pods) and the Gauge Point distribution (which determines how much Grown Stalk accrues to each asset) in each Season.

Other changes in the audited diff add a conversion type that Chops Unripe assets deposited in the Silo and deposits the corresponding underlying asset, support for changing Temperature on a relative scale, and accounting for the Deposited BDV of deposits not yet migrated to Silo V3.

9 Attestation

Cyfrin attests that it performed a security review of the Beanstalk BIP-39 code provided by Beanstalk Farms.

The repositories, commits, review period and findings are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-05-02-cyfrin-beanstalk-bip-39-v1-2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Beanstalk Farms has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=9f2c5afa925d477e1fcdb092c3c1b1298d420117b218a150a1ef9627116b37b0
F=2024-05-02-cyfrin-beanstalk-bip-39-v1-2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.